

COS214 Practical Assignment 2

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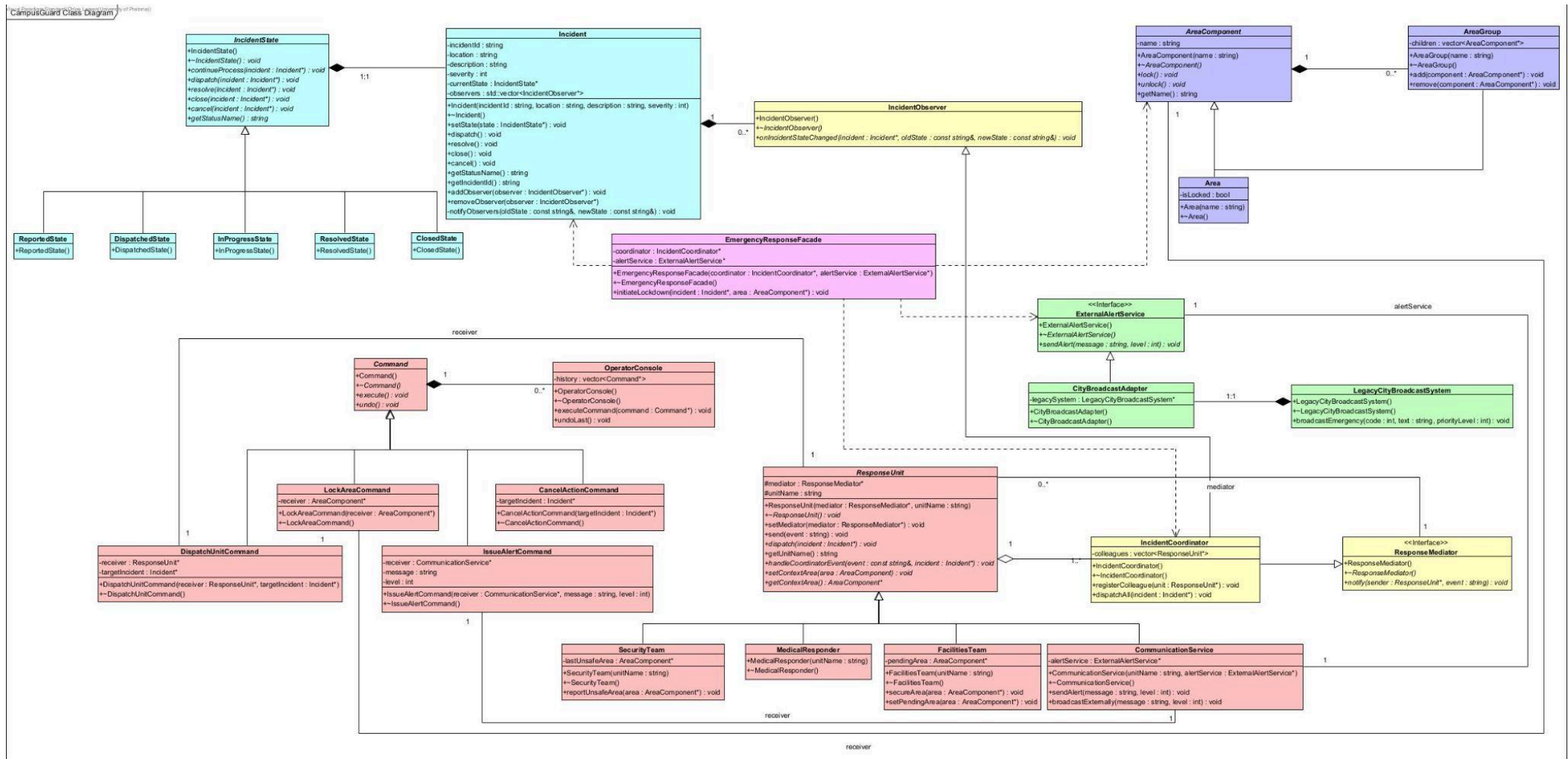
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Task 1

System Description

CampusGuard is an emergency-response coordination platform for a university campus that supports an incident from the moment it is reported until it is closed. Each incident is registered with an identifier, location, description and severity, and moves through the statuses Reported, Dispatched, In Progress, Resolved and Closed. An incident can be cancelled before it is resolved, in which case it moves straight to Closed, and any operation that is not valid in its current status is rejected with a message rather than silently ignored. Operators act through requests that are represented as objects and run from a single operator console: dispatching a response unit, locking an area, issuing an alert and cancelling an incident, with the most recent request able to be undone. The security team, medical responder, facilities team and communication service do not communicate with each other directly but through a central incident coordinator; for example, when the security team reports an area as unsafe, the coordinator has the facilities team secure the area and the communication service broadcasts an advisory. The campus is modelled as a hierarchy of buildings and rooms, so locking a building locks every room inside it, while a single room can still be locked on its own. External alerts, whether raised by the communication service or by the lockdown operation, are sent to the city's legacy emergency broadcast system, whose interface differs from the one CampusGuard uses, so an adapter translates each alert into the legacy format. Finally, a single lockdown operation gives the operator a high-level entry point that locks an area, dispatches all registered response units, advances the incident's status and raises an external alert in one controlled sequence, while each of those operations remains available on its own.

CampusGuard Class Diagram



GoF Participant Mapping

Command

Command	Command
ConcreteCommand	DispatchUnitCommand, LockAreaCommand, IssueAlertCommand, CancelActionCommand
Invoker	OperatorConsole
Receiver	ResponseUnit, AreaComponent, CommunicationService, Incident

Mediator

Mediator	ResponseMediator
ConcreteMediator	IncidentCoordinator
Colleague	ResponseUnit
ConcreteColleague	SecurityTeam, MedicalResponder, FacilitiesTeam, CommunicationService

Adapter

Adaptee	LegacyCityBroadcastSystem
Target	ExternalAlertService
Adapter	CityBroadcastAdapter
Client	CommunicationService, EmergencyResponseFacade

Facade

Facade	EmergencyResponseFacade
Subsystem classes	AreaComponent, IncidentCoordinator, Incident, ExternalAlertService

Composite

Component	AreaComponent
Leaf	Area
Composite	AreaGroup
Client	EmergencyResponseFacade, LockAreaCommand,

	FacilitiesTeam
--	----------------

State

Context	Incident
State	IncidentState
ConcreteState	ReportedState, DispatchedState, InProgressState, ResolvedState, ClosedState

Justification

State (Incident status)

An incident's status decides what can happen to it: it can only be cancelled early on, and it can only be closed once it has been resolved. Without a clear structure, those rules would end up scattered across the code and be easy to break. The simple option would be a status field checked with “if/else” chains everywhere, which the practical doesn't allow, and every new status would mean touching all of that code. With State, each status is its own class that knows only what is allowed from that point, rejects everything else, and moves the incident to the next status. Adding a status means adding a class and changing nothing else.

In the workflow, the incident hands every request to whichever status it is currently in. In Scenario 1 the lockdown moves it forward, and in Scenario 2 an attempt to resolve a newly reported incident is rejected, which shows our failure case.

Composite (Campus areas)

A campus is a hierarchy of campus, buildings and rooms, and the operator needs to lock any level in the same way. Locking a room and locking a whole building should be the same request, with the building request reaching every room inside it. The simple option would be separate room and building types, with code that checks which one it has and loops through the rooms itself, and every place that locks an area would have to repeat that check. With Composite, rooms and buildings share one interface, so the caller doesn't need to know which it is dealing with, and a building simply passes the request on to what it contains.

In the workflow, the lockdown in Scenario 1 locks the whole campus in one request and every room is locked. In Scenario 2 a command locks just one lab through the same interface.

Design and ownership rational

Design rationale

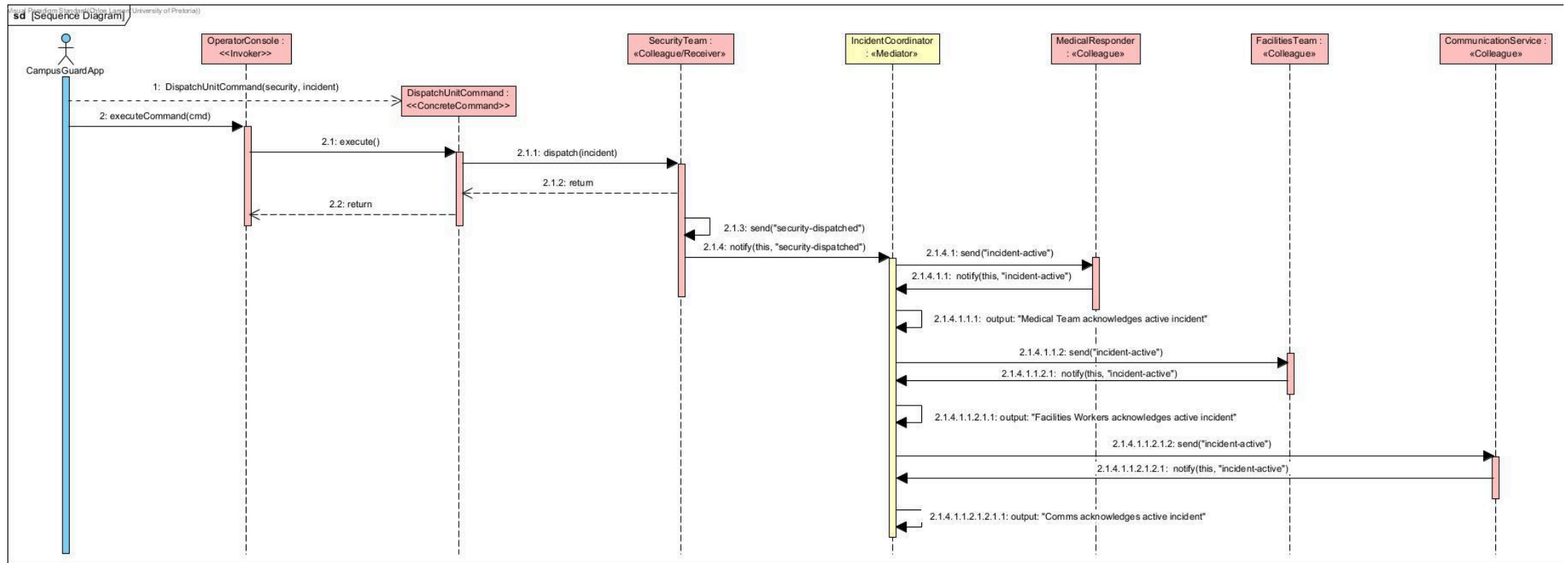
We kept the design as small as the requirements allowed: one class for each real responsibility, and a pattern only where it removed a specific problem. Operator actions are objects, so a request is separate from the code that carries it out and the latest one can be undone. The response units talk through a coordinator, so no unit needs to know about the others. The legacy city system sits behind an adapter, so the rest of the application only sees the interface it expects. The lockdown is a single entry point, so callers don't need to know the order of the steps, and each step is still usable on its own. State and Composite follow the reasoning above. Where one object just needs to use another, it keeps a reference and does not take responsibility for it, which keeps coupling low.

Ownership

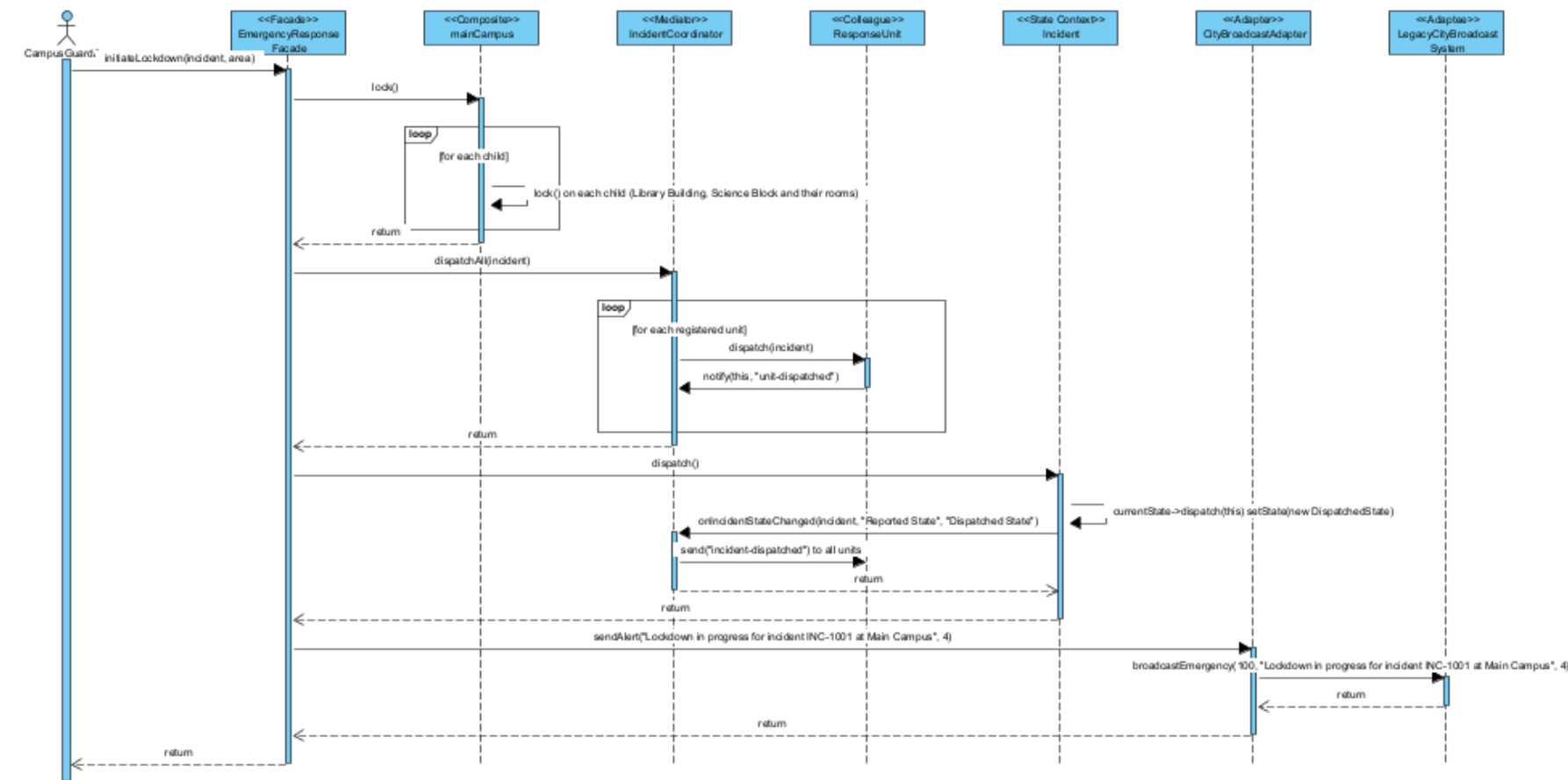
The rule is that whoever creates an object, or is handed it to look after, deletes it, and everything else only borrows it. The application object at the top owns the coordinator, the adapter, the console, the four units, the campus tree and the current incident. It deletes them in its destructor, removing the units before the coordinator and adapter they refer to. Three objects own things of their own. An incident owns its current status and replaces it on every transition. A building owns everything added to it, so deleting the campus frees the whole tree. The operator console owns the commands it is given and deletes them when it is destroyed or when they are undone. The adapter also creates and deletes the legacy system it wraps. Everything else is borrowed: the coordinator and its units refer to each other without owning, commands refer to their receivers, the communication service refers to the adapter, and the facade owns nothing. Every base class used polymorphically has a virtual destructor, so deleting through the base type cleans up the full object.

Task 4

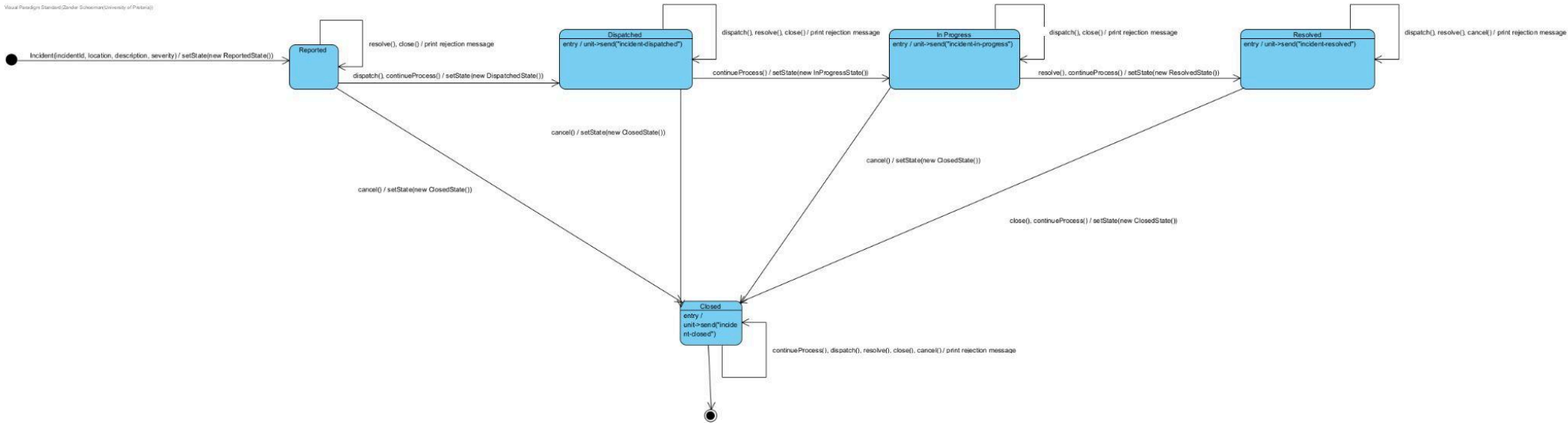
Sequence Diagram - Command and Mediator



Sequence Diagram - Facade



Additional behavioural diagram - State Diagram



Task 5

GDB Evidence

```
gdb ./campusGuard
```

```
Reading symbols from ./campusGuard...
```

```
(gdb) run
```

```
Starting program: /mnt/c/Users/chloe/OneDrive - University of Pretoria/Second Year/COS  
214/Practical/COS214-Prac5/campusGuard
```

```
=====
```

CampusGuard Setup

```
=====
```

New incident coordinator has been created

New city broadcast adapter has been created

New security team Security Bros has been created

New medical responder Medical Team has been created

New facilities team Facilities Workers has been created

New communication service Comms has been created

Adding Security Bros unit to Incident Coordinator

Adding Medical Team unit to Incident Coordinator

Adding Facilities Workers unit to Incident Coordinator

Adding Comms unit to Incident Coordinator

New operator console has been created

```
=====
```

Campus composite tree setup

```
=====
```

Main Campus (Area Group) has been created

Library Building (Area Group) has been created

Reading Room (Area) has been created and is unlocked

Reading Room has been added to Library Building

Archive (Area) has been created and is unlocked

Archive has been added to Library Building

Server Room (Area) has been created and is unlocked

Server Room has been added to Library Building

Science Block (Area Group) has been created

Chem Lab 1 (Area) has been created and is unlocked

Chem Lab 1 has been added to Science Block

Chem Lab 2 (Area) has been created and is unlocked

Chem Lab 2 has been added to Science Block

Library Building has been added to Main Campus

Science Block has been added to Main Campus

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SCENARIO 1: Fire in the Library

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Incident (INC-1001) with a severity level of 4 at Library Building - Smoke detected in Reading Room has been created

Dispatch Unit Command capabilities have been added to Incident(INC-1001) for Response unit Security Bros

Units to incident (INC-1001) through response unit Security Bros have been dispatched Security Team Security Bros responding to Incident(INC-1001)

Coordinator event security-dispatched From Security Bros

Coordinator event incident-active From Medical Team

Coordinator Medical Team acknowledges active incident

Coordinator event incident-active From Facilities Workers

Coordinator Facilities Workers acknowledges active incident

Coordinator event incident-active From Comms

Coordinator Comms acknowledges active incident

Dispatch Unit Command capabilities have been added to Incident(INC-1001) for Response unit Facilities Workers

Units to incident (INC-1001) through response unit Facilities Workers have been dispatched

Facilities Team Facilities Workers responding to Incident(INC-1001)

Coordinator event facilities-dispatched From Facilities Workers

Coordinator event facilities-on-scene From Security Bros

Coordinator event facilities-on-scene From Medical Team

Coordinator event facilities-on-scene From Comms

Security Team Security Bros has reported at area Main Campus is unsafe

Coordinator event area-unsafe From Security Bros

===== LOCKDOWN INITIATED for incident INC-1001 (Reported State) =====

Facade step 1: locking area Main Campus

Main Campus is beginning the lock process

Library Building is beginning the lock process

Reading Room has been locked

Archive has been locked

Server Room has been locked

Science Block is beginning the lock process

Chem Lab 1 has been locked

Chem Lab 2 has been locked

Facade step 2: dispatching response units via coordinator

Coordinator dispatching all registered units for incident (INC-1001)

Security Team Security Bros responding to Incident(INC-1001)

Coordinator event security-dispatched From Security Bros

Coordinator event incident-active From Medical Team
Coordinator Medical Team acknowledges active incident
Coordinator event incident-active From Facilities Workers
Coordinator Facilities Workers acknowledges active incident
Coordinator event incident-active From Comms
Coordinator Comms acknowledges active incident
Medical Responder Medical Team responding to Incident(INC-1001)
Coordinator event medical-dispatched From Medical Team
Coordinator event medical-on-scene From Security Bros
Coordinator event medical-on-scene From Facilities Workers
Coordinator event medical-on-scene From Comms
Facilities Team Facilities Workers responding to Incident(INC-1001)
Coordinator event facilities-dispatched From Facilities Workers
Coordinator event facilities-on-scene From Security Bros
Coordinator event facilities-on-scene From Medical Team
Coordinator event facilities-on-scene From Comms
Communication Service Comms has no dispatch role for Incident(INC-1001)

Facade step 3: advancing incident state

Coordinator Incident (INC-1001) transitioned: Reported State -> Dispatched State
Coordinator event incident-dispatched From Security Bros
Coordinator event incident-dispatched From Medical Team
Coordinator event incident-dispatched From Facilities Workers
Coordinator event incident-dispatched From Comms

State: Reported State -> Dispatched State

Incident (INC-1001) has been dispatched

Facade step 4: raising external alert

CITY LEGACY BROADCAST

code = 100

priority = 4

text:Lockdown in progress for incident INC-1001 at Main Campus

===== LOCKDOWN WORKFLOW COMPLETE =====

Issue Alert Command capabilities have been added to communication service Comms with a message of "Evacuate Library Building immediately" at level 5

Message:Evacuate Library Building immediately

Level: 5

is being sent to communication service Comms

Communication Service Comms issuing on-campus alert: Evacuate Library Building immediately (level 5)

CITY LEGACY BROADCAST

code = 100

priority = 5

text:Evacuate Library Building immediately

Coordinator Incident (INC-1001) transitioned: Dispatched State -> In Progress State

Coordinator event incident-in-progress From Security Bros

Coordinator event incident-in-progress From Medical Team

Coordinator event incident-in-progress From Facilities Workers

Coordinator event incident-in-progress From Comms
State: Dispatched State -> In Progress State
Incident (INC-1001) is now in progress
Coordinator Incident (INC-1001) transitioned: In Progress State -> Resolve State
Coordinator event incident-resolved From Security Bros
Coordinator event incident-resolved From Medical Team
Coordinator event incident-resolved From Facilities Workers
Coordinator event incident-resolved From Comms
State: In Progress State -> Resolve State
Incident (INC-1001) is being resolved
Coordinator Incident (INC-1001) transitioned: Resolve State -> Closed State
Coordinator event incident-closed From Security Bros
Coordinator event incident-closed From Medical Team
Coordinator event incident-closed From Facilities Workers
Coordinator event incident-closed From Comms
State: Resolve State -> Closed State
Incident (INC-1001) has been closed
Messages broadcast cannot be unsent

=====

SCENARIO 1: COMPLETE

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SCENARIO 2: Medical Emergency

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Incident (INC-2002) with a severity level of 3 at Science Block - Student collapsed in Chem Lab 1 has been created

Program received signal SIGSEGV, Segmentation fault.
0x0000555555561cd4 in Incident::setState (this=0x55555558c060, state=0x55555558c140) at src/Incident.cpp:24
24 std::string oldName = currentState ? currentState->getStatusName() :
"<none>";
(gdb) bt
#0 0x0000555555561cd4 in Incident::setState (this=0x55555558c060,
state=0x55555558c140) at src/Incident.cpp:24
#1 0x0000555555561b57 in Incident::Incident (this=0x55555558c060,
incidentId="INC-2002", location="Science Block",
description="Student collapsed in Chem Lab 1", severity=3) at src/Incident.cpp:7
#2 0x000055555556cf73 in CampusGuardApp::scenario2 (this=0x55555558b2b0) at
main.cpp:180
#3 0x000055555556d4bc in CampusGuardApp::run (this=0x55555558b2b0) at main.cpp:200
#4 0x000055555556b802 in main () at main.cpp:207
(gdb) info locals
oldName = ""
newName = "\\340\\326\\377\\377\\377\\177\\000"

```

(gdb) frame 0
#0 0x00005555555561cd4 in Incident::setState (this=0x55555558c060,
state=0x55555558c140) at src/Incident.cpp:24
24         std::string oldName = currentState ? currentState->getStatusName() :
"<none>";
(gdb) list
19         void Incident::setState(IncidentState *state)
20         {
21             if (currentState == state || state == nullptr)
22                 return;
23
24             std::string oldName = currentState ? currentState->getStatusName() :
"<none>";
25
26             if(currentState)
27                 delete currentState;
28             currentState = state;
(gdb) q

```

I was getting a segfault in the setState function as I was using a currentState which has not been defined yet which was causing the seg fault. So the fix was just adding currentState(nullptr) to the constructor for the Incident class.

Valgrind Evidence

```

make valgrind
valgrind --leak-check=full --show-leak-kinds=all --track-origins=yes ./campusGuard
==40921== Memcheck, a memory error detector
==40921== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==40921== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==40921== Command: ./campusGuard
==40921==

```

CampusGuard Setup

```

New incident coordinator has been created
New city broadcast adapter has been created
New security team Security Bros has been created
New medical responder Medical Team has been created
New facilities team Facilities Workers has been created
New communication service Comms has been created
Adding Security Bros unit to Incident Coordinator
Adding Medical Team unit to Incident Coordinator
Adding Facilities Workers unit to Incident Coordinator

```

Adding Comms unit to Incident Coordinator

New operator console has been created

Campus composite tree setup

Main Campus (Area Group) has been created
Library Building (Area Group) has been created
Reading Room (Area) has been created and is unlocked
Reading Room has been added to Library Building

Archive (Area) has been created and is unlocked
Archive has been added to Library Building

Server Room (Area) has been created and is unlocked
Server Room has been added to Library Building

Science Block (Area Group) has been created
Chem Lab 1 (Area) has been created and is unlocked
Chem Lab 1 has been added to Science Block

Chem Lab 2 (Area) has been created and is unlocked
Chem Lab 2 has been added to Science Block

Library Building has been added to Main Campus

Science Block has been added to Main Campus

SCENARIO 1: Fire in the Library

Incident (INC-1001) with a severity level of 4 at Library Building - Smoke detected in Reading Room has been created

Dispatch Unit Command capabilities have been added to Incident(INC-1001) for Response unit Security Bros

Units to incident (INC-1001) through response unit Security Bros have been dispatched
Security Team Security Bros responding to Incident(INC-1001)

Coordinator event security-dispatched From Security Bros

Coordinator event incident-active From Medical Team

Coordinator Medical Team acknowledges active incident

Coordinator event incident-active From Facilities Workers

Coordinator Facilities Workers acknowledges active incident

Coordinator event incident-active From Comms

Coordinator Comms acknowledges active incident

Dispatch Unit Command capabilities have been added to Incident(INC-1001) for Response unit Facilities Workers

Units to incident (INC-1001) through response unit Facilities Workers have been dispatched

Facilities Team Facilities Workers responding to Incident(INC-1001)

Coordinator event facilities-dispatched From Facilities Workers

Coordinator event facilities-on-scene From Security Bros

Coordinator event facilities-on-scene From Medical Team

Coordinator event facilities-on-scene From Comms

Security Team Security Bros has reported at area Main Campus is unsafe

Coordinator event area-unsafe From Security Bros

===== LOCKDOWN INITIATED for incident INC-1001 (Reported State) =====

Facade step 1: locking area Main Campus

Main Campus is beginning the lock process

Library Building is beginning the lock process

Reading Room has been locked

Archive has been locked

Server Room has been locked

Science Block is beginning the lock process

Chem Lab 1 has been locked

Chem Lab 2 has been locked

Facade step 2: dispatching response units via coordinator

Coordinator dispatching all registered units for incident (INC-1001)

Security Team Security Bros responding to Incident(INC-1001)

Coordinator event security-dispatched From Security Bros

Coordinator event incident-active From Medical Team

Coordinator Medical Team acknowledges active incident

Coordinator event incident-active From Facilities Workers

Coordinator Facilities Workers acknowledges active incident

Coordinator event incident-active From Comms

Coordinator Comms acknowledges active incident

Medical Responder Medical Team responding to Incident(INC-1001)

Coordinator event medical-dispatched From Medical Team

Coordinator event medical-on-scene From Security Bros

Coordinator event medical-on-scene From Facilities Workers

Coordinator event medical-on-scene From Comms

Facilities Team Facilities Workers responding to Incident(INC-1001)

Coordinator event facilities-dispatched From Facilities Workers

Coordinator event facilities-on-scene From Security Bros

Coordinator event facilities-on-scene From Medical Team

Coordinator event facilities-on-scene From Comms

Communication Service Comms has no dispatch role for Incident(INC-1001)

Facade step 3: advancing incident state

Coordinator Incident (INC-1001) transitioned: Reported State -> Dispatched State

Coordinator event incident-dispatched From Security Bros

Coordinator event incident-dispatched From Medical Team

Coordinator event incident-dispatched From Facilities Workers

Coordinator event incident-dispatched From Comms

State: Reported State -> Dispatched State

Incident (INC-1001) has been dispatched

Facade step 4: raising external alert

CITY LEGACY BROADCAST

code = 100

priority = 4

text:Lockdown in progress for incident INC-1001 at Main Campus

===== LOCKDOWN WORKFLOW COMPLETE =====

Issue Alert Command capabilities have been added to communication service Comms with a message of "Evacuate Library Building immediately" at level 5

Message:Evacuate Library Building immediately

Level: 5

is being sent to communication service Comms

Communication Service Comms issuing on-campus alert: Evacuate Library Building immediately (level 5)

CITY LEGACY BROADCAST

code = 100

priority = 5

text:Evacuate Library Building immediately

Coordinator Incident (INC-1001) transitioned: Dispatched State -> In Progress State

Coordinator event incident-in-progress From Security Bros

Coordinator event incident-in-progress From Medical Team

Coordinator event incident-in-progress From Facilities Workers

Coordinator event incident-in-progress From Comms

State: Dispatched State -> In Progress State

Incident (INC-1001) is now in progress

Coordinator Incident (INC-1001) transitioned: In Progress State -> Resolve State

Coordinator event incident-resolved From Security Bros

Coordinator event incident-resolved From Medical Team

Coordinator event incident-resolved From Facilities Workers

Coordinator event incident-resolved From Comms

State: In Progress State -> Resolve State

Incident (INC-1001) is being resolved

Coordinator Incident (INC-1001) transitioned: Resolve State -> Closed State

Coordinator event incident-closed From Security Bros

Coordinator event incident-closed From Medical Team

Coordinator event incident-closed From Facilities Workers

Coordinator event incident-closed From Comms

State: Resolve State -> Closed State

Incident (INC-1001) has been closed

Messages broadcast cannot be unsent

=====

SCENARIO 1: COMPLETE

=====

=====

SCENARIO 2: Medical Emergency

=====

Incident (INC-2002) with a severity level of 3 at Science Block - Student collapsed in Chem Lab 1 has been created

State: Reported State

Incident (INC-2002) cannot be resolved

Dispatch Unit Command capabilities have been added to Incident(INC-2002) for Response unit Medical Team

Units to incident (INC-2002) through response unit Medical Team have been dispatched

Medical Responder Medical Team responding to Incident(INC-2002)

Coordinator event medical-dispatched From Medical Team

Coordinator event medical-on-scene From Security Bros

Coordinator event medical-on-scene From Facilities Workers

Coordinator event medical-on-scene From Comms

Chem Lab 1 (Area) has been created and is unlocked

Lock Area Command capabilities have been added to Chem Lab 1

Chem Lab 1 is getting locked:

Chem Lab 1 has been locked

Coordinator Incident (INC-2002) transitioned: Reported State -> Dispatched State

Coordinator event incident-dispatched From Security Bros

Coordinator event incident-dispatched From Medical Team

Coordinator event incident-dispatched From Facilities Workers

Coordinator event incident-dispatched From Comms

State: Reported State -> Dispatched State

Incident (INC-2002) has been dispatched

Cancel Action Command capabilities have been added to Incident(INC-2002)

Incident (INC-2002) is being canceled

Coordinator Incident (INC-2002) transitioned: Dispatched State -> Closed State

Coordinator event incident-closed From Security Bros

Coordinator event incident-closed From Medical Team

Coordinator event incident-closed From Facilities Workers

Coordinator event incident-closed From Comms

State: Dispatched State -> Closed State

Incident (INC-2002) has been canceled

=====

SCENARIO 2: COMPLETE

=====

==40921==

==40921== HEAP SUMMARY:

==40921== in use at exit: 0 bytes in 0 blocks

==40921== total heap usage: 207 allocs, 207 frees, 81,211 bytes allocated

==40921==

==40921== All heap blocks were freed -- no leaks are possible

==40921==

==40921== For lists of detected and suppressed errors, rerun with: -s

==40921== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)

Docker run instructions

From the repository root, in a clean shell:

```
docker compose up --build
```

This command:

1. Builds the image from the Dockerfile (installs the toolchain, runs `make clean && make`).
2. Creates the `campusguard-demo` container.
3. Runs `./campusGuard` non-interactively — the two end-to-end scenarios execute to completion with no manual input.
4. The container exits when the application returns from `main()`.

GitGub

<https://github.com/Chloe-Larsen/COS214-Prac5>

Collaboration Reflection

Our team of three worked through a shared GitHub repository (Chloe-Larsen/COS214-Prac5), using separate branches and pull requests so that work could be reviewed before it reached `main`. Chloe implemented the core code base (PR #37) and maintained the Visual Paradigm project. Zander worked on [his part, e.g. the README, Docker setup] and Lebogang wrote the documentation and diagrams on the `lebo-github-docs` branch. We coordinated through a WhatsApp group, and when the implementation changed we updated the diagrams and written explanations to match. Merging the team's `main` into our own branches surfaced conflicts, which we resolved before pushing. One lesson was that the design documents drift from the code quickly, so they need to be reviewed after every significant implementation change.

Contribution Statement

Chloe Larsen: Implemented the class hierarchy and patterns (PR #37) and Command/Mediator sequence diagram.

Zander Schoeman: Completed task 1, the additional behavioural diagram which is a state diagram and the Reame

Lebogang Nxumalo: Completed the Facade sequence diagram, the GitHub history documentation, and the collaboration reflection and contribution statement.